

Total no. of Pages: 2...

Roll no.

Degree: BTech. (ECE-AIML), Semester: IIIrd
END-SEMESTER THEORY EXAMINATION, NOV-DEC 2024

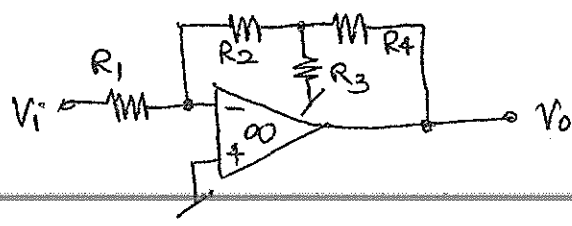
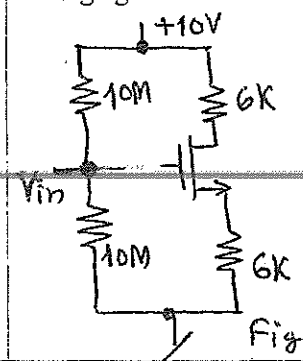
Course Code: EAEP-304

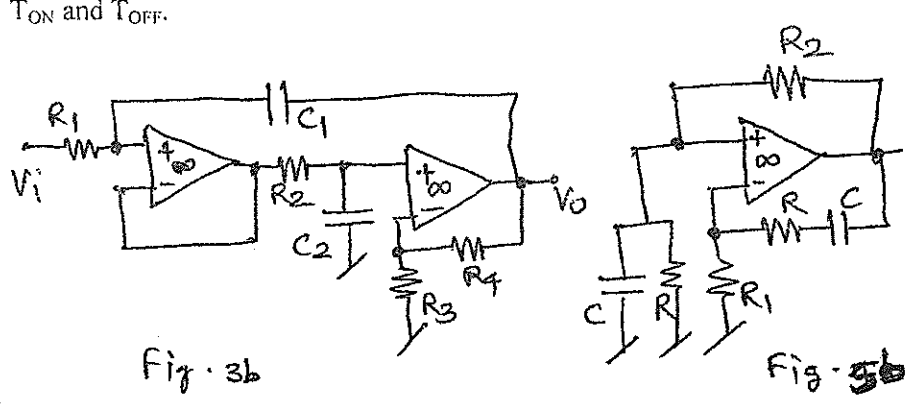
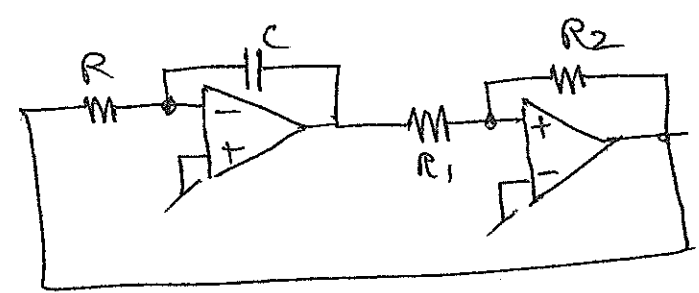
Course Title: Microelectronics Circuits and Applications

Time: 03 Hours

Max. Marks: 50

Note:- Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q.No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	With a neat circuit arrangement for output characteristics of an NPN BJT discuss the early effect and show by expression/characteristic how the same has been incorporated.	5	CO1
1b	For the circuit shown in Fig.1b determine the voltages at all nodes and current through all branches. Let $V_{in}=1V$, $k'_n \left(\frac{W}{L}\right) = 1mA/V^2$ and neglect channel width modulation.	5	CO1
1c	Draw the single stage Common-source and Common-gate amplifier circuit configurations and from their respective small-signal high-frequency equivalents discuss the miller-effect and thus, derive the upper-cut-off frequency ω_H .	5	CO1
Q 2	Attempt any 2 parts of the following		
2a	For a MOS differential pair with matched MOS and matched R_D derive the differential-mode gain. Draw the voltage transfer curve too.	5	CO2
2b	For a BJT differential pair derive the hyperbolic-relationship between differential output current (I_O) and its differential input voltage (V_{id}). Also draw the complete circuit diagram of the OTA.	5	CO2
2c	Discuss various effects of mismatches on the performance of differential pair.	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	Design the inverting amplifier given in Fig.3a for input impedance 1Mohm and voltage gain 10000.   Fig. 1b	5	CO3
3b	For the filter circuit given in Fig.3b derive the transfer function and deduce expression for ω_0 , Q_0 and Gain.	5	CO3

3c	From the general circuit diagram of the Antoniou's GIC derive the case of grounded inductor realisation and comment how its floating version can be made?	5	CO3
Q 4	Attempt any 2 parts of the following		
4a	For the circuit given in Fig.4a discuss its functionality and derive expression for the wave generated.	5	CO4
4b	Draw an OPAMP-based monostable multivibrator and discuss its operation with necessary derivations and waveforms.	5	CO4
4c	Draw a neat internal diagram of 555 timer IC and, additionally, with appropriate external circuit elements explain its astable operation. Also derive expressions for T_{ON} and T_{OFF} .	5	CO4
	 Fig. 3b Fig. 5b		
Q 5	Attempt any 2 parts of the following		
5a	Draw the block diagram of a Voltage-Shunt Feedback and deduce the expressions for R_{if} and R_{of} of the feedback amplifier in relation to R_i and R_o of the Amplifier without feedback.	5	CO5
5b	For the circuit given in Fig.5b analyse and test for its possibility of being an oscillator deduce expressions for its CO and FO.	5	CO5
5c	Draw the circuit diagram of a BJT-based Class-B push-pull configuration and derive expression for its efficiency, also discuss the harmonic distortion aspect of the same.	5	CO5
	 Fig-4a		